

3.2.

$$\langle f, g \rangle = \int_1^5 f(x)g(x)dx, \quad \forall \dots \text{v.p. spoj. } \mathbb{R} \text{ fceí s } \mathcal{D}_f = [1, 5]$$

a) 1. $\text{Span} \{1, x, x^2\}, \quad \vec{v}_1 = 1 \quad \vec{v}_2 = x \quad \vec{v}_3 = x^2$

$$\vec{u}_1 = \frac{\vec{v}_1}{\|\vec{v}_1\|} = \frac{1}{\sqrt{\int_1^5 1^2 dx}} = \frac{1}{\sqrt{4}} = \frac{1}{2}$$

$$\vec{w}_2 = \langle \vec{u}_1, \vec{v}_2 \rangle \vec{u}_1 = \left(\int_1^5 \frac{1}{2} \cdot x dx \right) \cdot \frac{1}{2} = \frac{6}{2} = 3$$

$$\vec{u}_2 = \frac{\vec{v}_2 - \vec{w}_2}{\|\vec{v}_2 - \vec{w}_2\|} = \frac{x-3}{\sqrt{\int_1^5 (x-3)^2 dx}} = \frac{x-3}{\sqrt{\frac{16}{3}}} = \frac{x-3}{4} = \frac{\sqrt{3}(x-3)}{4} = \frac{\sqrt{3}}{4}x - \frac{3\sqrt{3}}{4}$$

$$\vec{w}_3 = \langle \vec{u}_1, \vec{v}_3 \rangle \vec{u}_1 + \langle \vec{u}_2, \vec{v}_3 \rangle \vec{u}_2 = \left(\int_1^5 \frac{1}{2} x^2 dx \right) \cdot \frac{1}{2} + \left(\int_1^5 \left(\frac{\sqrt{3}}{4}x - \frac{3\sqrt{3}}{4} \right) \cdot x^2 dx \right) \cdot \left(\frac{\sqrt{3}}{4}x - \frac{3\sqrt{3}}{4} \right) = \frac{31}{3} + 6x - 18 = 6x - \frac{23}{3}$$

$$\vec{u}_3 = \frac{\vec{v}_3 - \vec{w}_3}{\|\vec{v}_3 - \vec{w}_3\|} = \frac{x^2 - 6x + \frac{23}{3}}{\sqrt{\int_1^5 (x^2 - 6x + \frac{23}{3})^2 dx}} = \frac{x^2 - 6x + \frac{23}{3}}{\frac{16\sqrt{5}}{15}} = \frac{\sqrt{5}}{16} (3x^2 - 18x + 23)$$

orton. báze B tedy je: $B = \left(\frac{1}{2}, \frac{\sqrt{3}}{4}x - \frac{3\sqrt{3}}{4}, \frac{\sqrt{5}}{16} (3x^2 - 18x + 23) \right)$

b) podle tvrzení 8.54

řekněme že vektor $\vec{p}$ je ortogon. projekcí $\vec{S} = \sin x$ na podprostor V s bází B pokud platí

$$\vec{p} = \langle \vec{u}_1, \vec{S} \rangle \vec{u}_1 + \langle \vec{u}_2, \vec{S} \rangle \vec{u}_2 + \langle \vec{u}_3, \vec{S} \rangle \vec{u}_3 =$$

$$= \left(\int_1^5 \frac{1}{2} \cdot \sin x dx \right) \cdot \frac{1}{2} + \left(\int_1^5 \left(\frac{\sqrt{3}}{4}x - \frac{3\sqrt{3}}{4} \right) \cdot \sin x \right) \cdot \left(\frac{\sqrt{3}}{4}x - \frac{3\sqrt{3}}{4} \right) + \left(\int_1^5 \left(\frac{\sqrt{5}}{16} (3x^2 - 18x + 23) \right) \cdot \sin x \right) \cdot \left(\frac{\sqrt{5}}{16} (3x^2 - 18x + 23) \right) =$$

$$\approx 0,064 + 1,9395 - 0,6465x - 0,0525x^2 + 0,315x - 0,4024 = -0,0525x^2 - 0,3315x + 1,6011$$

c) norma vzdálosti $\|\sin x - (-0,0525x^2 - 0,3315x + 1,6011)\| =$

$$= \sqrt{\int_1^5 (\sin x - (-0,0525x^2 - 0,3315x + 1,6011))^2 dx} \approx 0,3188$$